

AI FOR AERODYNAMICS

V Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE16	PCC	L	T	P	C	CIA	SEE	Total
		3	1	-	4	40	60	100
UNIT-I	INTRODUCTION AI							
Introduction, history, intelligent systems, foundations of AI, Expert System applications, tic-tac-toe –game playing, current trends in AI. Artificial Intelligence and Aviation Applications								
UNIT-II	MACHINE LEARNING							
Introduction, Well-Posed Learning Problems, Perspectives and Issues in Machine Learning, Supervised learning, unsupervised learning,.								
UNIT-III	FUZZY LOGIC and NEURAL NETWORKS							
Introduction, neural network representation, models of a neuron, network architecture, multi-layer networks, Fuzzy Logic-Integrated AI Systems-Design Considerations for Avionic Applications- biological and artificial neurons								
UNIT-IV	AERODYNAMIC DESIGN OF AIRFOILS WITH AI TOOLS							
Conventional airfoil design-estimation aerodynamic coefficients-supercritical airfoil design- estimation aerodynamic coefficients-shape optimization of airfoil NACA airfoils- symmetric –cambered ,customized airfoils								
UNIT-V	Aerodynamic design of aircraft components with AI tools							
Introduction-React angular wings- tapered wings-Delta wings-horizontal stabilizer-vertical stabilizer-fuselage-shape optimization of wings								
Text Books:								
1.Ethem Alpaydın ,Introduction to Machine Learning, The MIT Press Cambridge, Massachusetts London, England 2.Stuart J. Russell and Peter Norvig, Artificial Intelligence A Modern Approach Third Edition								
Reference Books:								
1.FAA doc ‘ Artificial Intelligence With Applications for Aircraft’								
Course Outcomes : Students should be able to: At the end of the course the students are able to 1.Illustrate the expert systems used in aerodynamics 2. Discuss the issues in machine learning 3.explain the fuzzy logic applications in aerodynamics 4.Formulate algorithm for aerodynamic efficiency of airfoils 5. Demonstrate the shape optimization of aircraft components								